Addinext Platform – Real-time Decentralized Architecture

WORKFLOW LAYERS ← INTER-LAYER FLOW (Linked via Patient ID) → SERVICE / API LAYER | DECENTRALIZED DATABASES

PATIENT INTAKE LAYER

Patient Registration → Demographics & Medical History → Clinical Assessment → Measurements & Diagnosis → Device Prescription (Type, Side, Priority)

Patient Service API
- Data Validation
- Business Logic
- API Endpoints
- Auto-save Enabled

Real-time Write Auto-save →
← Fetch / Read Live Sync

Patient Records Database (De-centralized)

Linked via Patient ID

LAYER 2 – 3D SCANNING LAYER

3D Scan Capture (Creality Raptor / NIR) → STL Export → File Validation (Face Count, Integrity Check) → Upload to Platform (Linked to Patient Record)

Scan Service API
- File Validation
- Metadata Check
- API Endpoints
- Auto-save Enabled

Real-time Write Auto-save →
← Fetch / Read Live Sync

Scan Files Database (De-centralized)

Linked via Patient ID

LAYER 3 – ORTHOPRINT MODULE

Clean (Fragment Removal, Spike Smoothing) | Adjust (Volume, Rings, Relief) | Align (PCA, Z-axis Orientation) | Trim (Rhino Split, 8-Phase Workflow) | Design (Shell, Wall Thickness, Decimation) | Validate (Watertight Mesh, Circumference Check) | Clinical Sign-off (Approval Gate) | Print Preperation (STL Export, Slicing Params, Material Spec: PETG / CF-PLA)

Design Service API
- Design Rules
- Validation Checks
- Version Control
- API Endpoints
- Auto-save Enabled

Real-time Write Auto-save →
← Fetch / Read Live Sync

Design Files Database (De-centralized)

Linked via Patient ID

LAYER 4 – MANUFACTURING LAYER

3D Printing (FDM / SLS Fabrication) → Post-processing (Support Removal, Finishing) → Quality Control (Fit Check, Tolerance Validation)

Manufacturing Service API
- Process Tracking
- QC Validation
- Job Status Update
- API Endpoints
- Auto-save Enabled

Real-time Write Auto-save →
← Fetch / Read Live Sync

Manufacturing Data Database (De-centralized)

Linked via Patient ID

LAYER 5 – DATABASE LAYER (DECENTRALIZED STORAGE)

Patient Records Database | Scan Files Database | Design Files Database | Manufacturing Data Database | Audit Logs & Traceability

Data Orchestration Layer
- Cross-Module Sync
- Data Integrity
- Access Control
- Audit & Logging

Global Capabilities
- ✓ Real-time Sync Across Modules
- ✓ Linked via Patient ID
- ✓ Role-based Access Control
- ✓ End-to-End Data Traceability
- ✓ Scalable & Decentralized

PLATFORM CAPABILITIES

- Patient ID Across All Layers
- Real-time Write Auto-save Enabled
- Live Sync Fetch / Read
- Instant Data Storage At Moment of Creation
- Decentralized & Scalable
- Secure, Compliant & Reliable